

Harvesting risk, not return

A governed systematic platform built on an uncomfortable finding: at the horizons and instruments tested, there was no exploitable edge in *predicting* returns. The edge that survives out-of-sample lives in the *risk structure*.

by **Carter Warrens** · Blaque Baux · 2026 · reference implementation in Julia

ABSTRACT

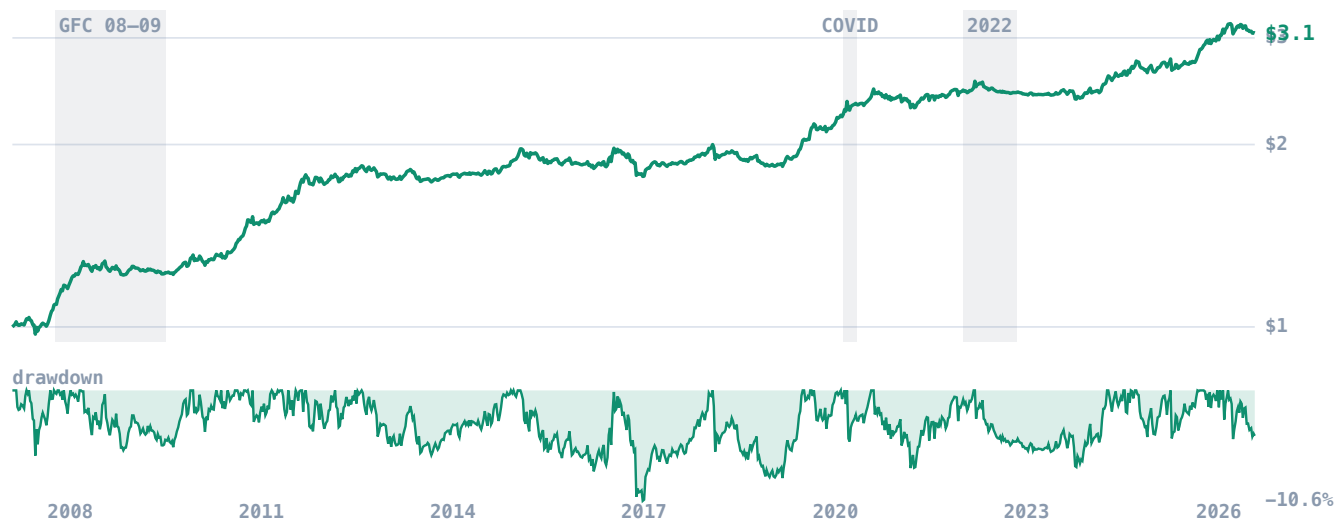
We describe a risk-premium harvesting strategy — the **spine** — that allocates by risk rather than return forecast: an inverse-volatility / equal-risk-contribution base blended with a 12-month time-series-momentum trend sleeve, each volatility-targeted on its own realized P&L, scaled by a drawdown regime brake, and rebalanced daily. On 2007–2026 (net of costs) it returns a Sharpe of ≈ 0.97 at a $\approx 11\%$ maximum drawdown, with the trend sleeve carrying the book through 2022. We motivate the design from a direct measurement — candidate return-prediction signals tested at $\approx 50\%$ accuracy (a coin flip) — and we quantify, rather than hide, the strategy's ceiling: this is single-digit-CAGR, ~ 1.0 -Sharpe money, and leverage to reach double digits buys proportional drawdown and is net-negative at today's financing rates.

1 The thesis

Most systematic strategies are sold on their ability to forecast. We tested that ability directly. On 60 days of 15-minute data across ~ 60 S&P names, two candidate alpha models — a three-factor cross-sectional model and a Bayesian Kalman/shrinkage engine — scored **50.9%** and **50.7%** directional accuracy, with information coefficients of **-0.004** and **+0.001**. Both are coin flips.

So we stopped trying to predict returns and built around what is measurable and persistent: the compensation paid for *bearing risk others avoid* — the diversification premium, and the trend (crisis-alpha) premium. Covariance is more forecastable than mean; the strategy trades on it.

FIGURE 1 – GROWTH OF \$1 AND DRAWDOWN, 2007–2026



The spine, unlevered, net of costs. \$1 → \$3.1 over ~19 years at ~6% CAGR; maximum drawdown **-10.6%** — no single crisis (GFC, COVID, 2022) worse than ~-7%. Shaded: the three crisis windows.

2 Construction

The book is two sleeves. Both are sized by an exponentially-weighted covariance estimate Σ (RiskMetrics, half-life 21), never by a return forecast.

Base sleeve — inverse-volatility weights harvest the diversification premium by putting risk, not dollars, on an equal footing (an equal-risk-contribution variant is also implemented):

$$w_i^{\text{base}} = \frac{1/\sigma_i}{\sum_j 1/\sigma_j}$$

Trend sleeve — a 12-month time-series-momentum sign, inverse-vol scaled into a long/short book. It goes *short* what is falling, which is why it made money in 2022:

$$s_i = \text{sign} \left(\prod_{u=t-251}^t (1 + r_u) - 1 \right), \quad w_i^{\text{trend}} = \frac{s_i / \sigma_i}{\sum_j |s_j / \sigma_j|}$$

Each sleeve is scaled to a target volatility using its *own realized P&L* volatility (a RiskMetrics recursion), capped — the load-bearing choice that keeps the trend hedge engaged in a crisis, when its P&L is smooth even as asset vols spike:

$$v_t^2 = \lambda v_{t-1}^2 + (1 - \lambda) p_t^2, \quad e_t = \min \left(\text{cap}, \frac{\sigma^*}{v_t \sqrt{\text{ppy}}} \right)$$

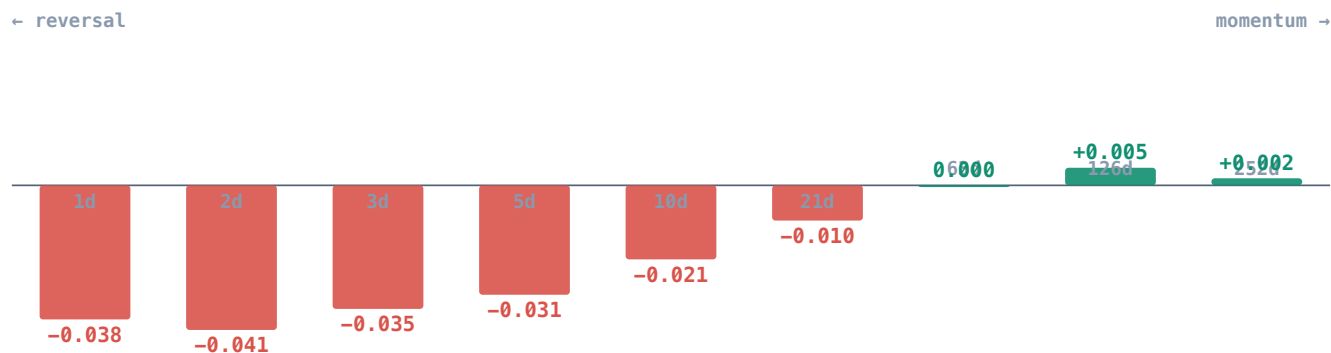
The vol-targeted sleeves are blended 50/50 and the whole book is scaled by a drawdown **regime brake** g_t that halves gross exposure when the market is >8% below its recent peak:

$$w_t = g_t \cdot \left(\theta e^{\text{base}} w^{\text{base}} + (1 - \theta) e^{\text{trend}} w^{\text{trend}} \right)$$

3 Why daily, and why twelve months

Rebalance cadence and lookback are not free parameters — they are dictated by where momentum actually exists. Measuring the information coefficient of a trailing move against the next day's return, by horizon, shows the market *reverses* at short horizons and only *trends* over months:

FIGURE 2 – MOMENTUM VS. REVERSAL BY HORIZON (IC, 2006–2026)

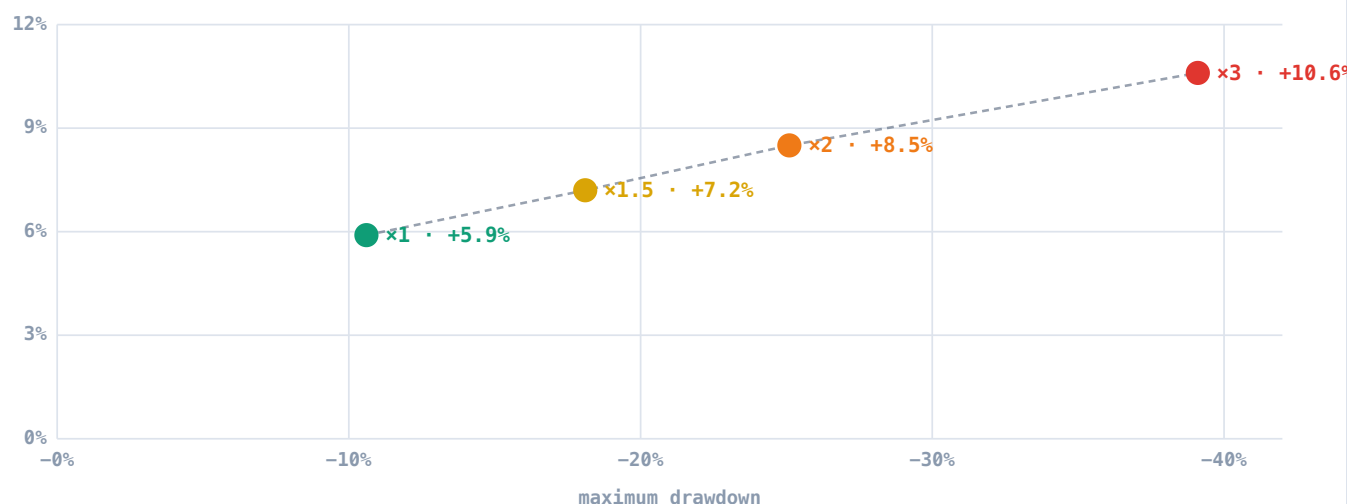


Trailing-return → next-day IC, pooled across assets. **Short horizons (1–63d) are reversal** (negative IC; 2-day is strongest); **momentum appears only at 126–252d**. This is why the trend sleeve uses a 12-month lookback and why "shorting a 2-day selloff" fights the strongest short-horizon force in the data. Reacting intraday, tested on 17,659 15-minute bars, added nothing — a diversified book absorbs intraday shocks internally.

4 The leverage trade-off, quantified

The honest ceiling. Leverage L on the strategy return, financing the borrowed portion at rate f , is $r_t^L = L r_t - (L-1)f/ppy$. Two facts follow, and neither is comfortable:

FIGURE 3 – RETURN VS. MAXIMUM DRAWDOWN BY LEVERAGE (2007–2026, 3% FINANCING)



Each point is the same strategy at a different dial setting. **Sharpe only falls with leverage** (financing is pure drag); reaching double-digit CAGR requires ~3x, which comes with a **–39% drawdown**. At today's ~6.5% margin the upper settings earn *less* than x1 — the ~6% return is below the cost of borrowing. There is no double-digit-at-low-risk point on this curve.

LEVERAGE	CAGR (3% FIN)	CAGR (6.5% FIN)	SHARPE	MAX DD
x1	+5.9%	+5.9%	0.97	-10.6%
x1.5	+7.2%	+5.4%	0.81	-18.1%
x2	+8.5%	+4.7%	0.73	-25.1%
x3	+10.6%	+3.2%	0.64	-39.1%

5 Governance, and honesty as a feature

Every order passes one governed path enforcing hard invariants — idempotency, per-pool budget and daily-loss limits, data-staleness, position reconciliation, a kill switch, and full fill lineage — beneath a pre-trade safety gate (drawdown and loss halts, gross-leverage and per-name caps). No language model is anywhere near the order path; the strategy is reproducible code.

Just as important is what we *do not* claim. The Bayesian alpha engine, the Monte-Carlo simulator, and a Sharpe-on-historical-means optimizer are all preserved but **archived**: the first measured as noise, the second is a risk calculator (it sizes a tail; it does not create return), and the third is the

Markowitz error-maximization trap. Sophisticated machinery packages and sizes a signal; it does not create one. That distinction is the whole paper.

6 Recent extensions — IN LIVE TESTING, NOT YET PART OF THE VALIDATED CORE ABOVE

Two changes are being trialled since the results above were established. Both are honest works in progress: validated only on a shorter **2016–2026** window (no 2008 crisis in it), so they lean as much on economic reasoning as on the backtest, and they graduate into the results above only if they hold up out-of-sample on live paper.

A sixth sleeve — DBA (agriculture). Of the diversifiers tested (international/EM equity, credit, REITs, silver), only agriculture proved genuinely uncorrelated to the existing five (0.44 vs 0.66–0.84 for the rest, which are equity/rate beta in disguise). Added to production, it lifted the backtested Sharpe from ≈ 0.94 to ≈ 1.04 at roughly flat drawdown. The economic case — weather- and supply-driven risk, orthogonal to financial markets — is the stronger justification.

A split-universe spine. The two sleeves want different universes: the long-only base wants risk-premium assets, while the long/short trend sleeve wants *breadth* — many independent markets so something is always trending. Giving trend its own broader (shortable) set — adding base-metals, silver, and USD/JPY — backtests at Sharpe ≈ 1.12 with lower volatility. It is running in a **live paper A/B** against the single-universe spine on a separate account; no funded capital is on it. If it wins on real out-of-sample data, it becomes the spine and this section moves up.

Full mathematics, code references, and citations: [docs/FINANCIAL_METHODS.md](#). Interactive leverage explorer: [docs/leverage_decision.html](#). Architecture: [docs/CANONICAL_ARCHITECTURE.md](#). Reference implementation (Julia): [the Blaque Baux repository](#).

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